Korean Lotto 6/45
Number Frequency, Prize Sharing and Trends

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Analysis of 1,245 draws (rounds 1–1245)
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Every set of 6 numbers has the same chance of being drawn.
This report shows which numbers appear most often, how prizes are shared among winners, and how these patterns have changed over time.

1. Data

Source: official Korean Lotto 6/45 results. 1,245 rounds. Main columns: six winning numbers, bonus, number of 1st-prize winners, prize per winning ticket (KRW). Amounts are shown in KRW billion (10^9 won).

Check	Result
Rounds	1,245 (no gaps, no duplicates)
Rounds with zero 1st-prize winners	14
Total 1st-prize winners	10,697
All numbers in 1–45, sorted	Yes
Missing values	0

Variable	Mean	Median	Min	Max
Winners per round	8.59	8	0	63
Prize per winner (bn KRW)	2.66	2.10	0.41	40.72
Total 1st-prize pool (bn KRW)	17.48	15.45	2.00	83.60

2. Which numbers appear most and least often?

Question. Over 1,245 draws, which numbers came out most frequently and which came out least?

What this table shows. Each draw picks 6 numbers. In 1,245 draws the expected count for any single number is 166. The table lists the ten most common and the ten least common numbers. A chi-square test checks whether the overall distribution differs from a uniform one.

Most frequent Least frequent

Rank	Number	Times drawn	Rank	Number	Times drawn
1	34	187	1	9	137
2	13	184	2	22	146
3	27	181	3	32	147
4	12	180	4	23	149
5	18	180	5	41	151
6	33	175	6	5	154
7	45	175	7	25	154
8	37	174	8	29	156
9	38	174	9	2	157
10	40	174	10	8	158

Expected frequency under equal chance = 166. Observed range: 137–187.

Analysis	Statistic	p-value
Chi-square test (all 45 frequencies vs uniform)	χ^2 = 48.3 (df=44)	0.30

The frequencies do not differ significantly from equal chance. The hottest and coldest numbers are still within normal random variation.

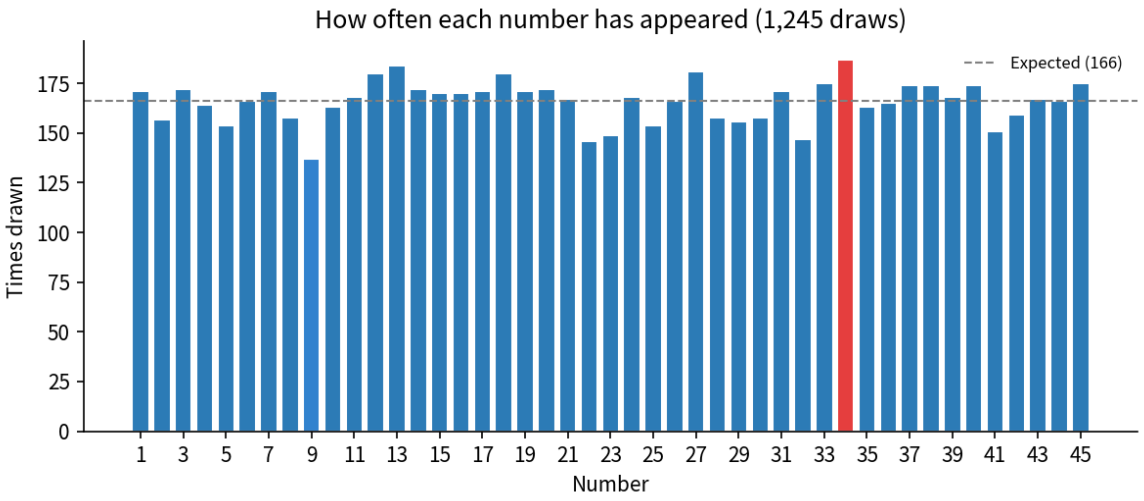


Figure 1. Frequency of each number 1–45. Red = most drawn (34), light blue = least drawn (9). Grey line = expected 166.

3. How does the prize change when more people win?

Question. When many tickets hit the same 1st prize, how much smaller is the amount each winner gets?

What this calculation does. Prize per winner P = total pool T / number of winners W. We fit a straight line on the log scale: ln P = a + b · ln W. The slope b tells how fast the prize falls. A slope of –1 means the total pool is fixed. A slope closer to zero means bigger winner counts come with bigger pools. Spearman rank correlation checks the same relationship without assuming a straight line.

Analysis	Statistic	p-value
OLS slope (ln prize ~ ln winners)	–0.683 (SE 0.014)	< 0.001
R^2 of the log-log fit	0.658	—
Spearman ρ (winners, prize)	–0.723	< 0.001
Spearman ρ (winners, total pool)	+0.578	< 0.001

Slope –0.683 means: if the number of winners doubles, the prize each person receives falls by about 38 %. The total pool itself grows when more people win, so the fall is less steep than –50 %.

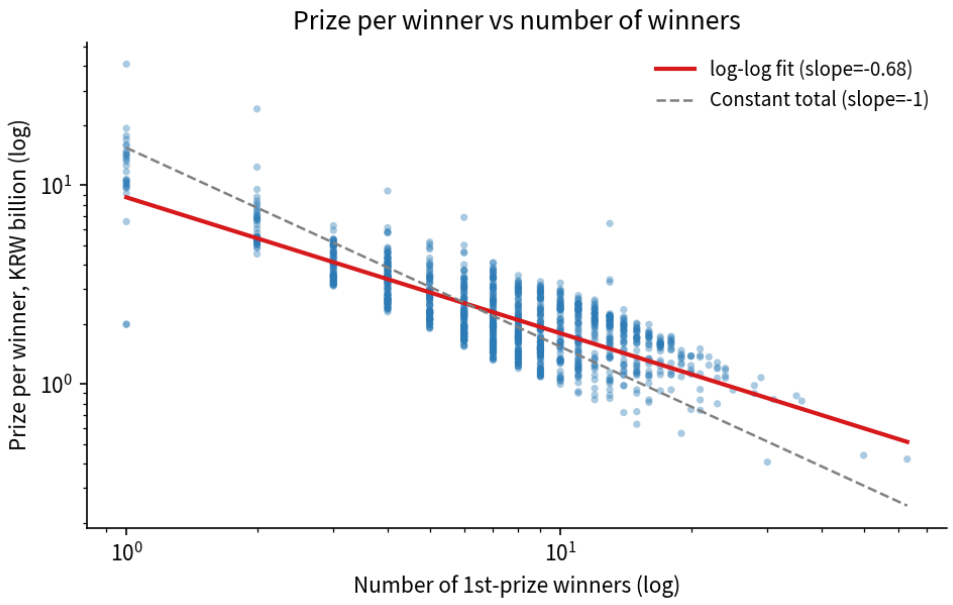


Figure 2. Prize per winner vs number of winners (both on log scale). Red line = fitted slope –0.68. Grey dashed line = fixed-pool slope –1.

Prize at different levels of sharing

Winners	Rounds	Mean prize	Median prize	Median total
1–2	68	9.34	7.39	11.90
3–4	177	3.76	3.57	13.00
5–7	333	2.43	2.23	13.54
8–10	297	1.95	1.79	15.37
11–15	249	1.78	1.80	23.73
16–25	98	1.36	1.37	26.41
26+	9	0.75	0.84	26.46

All amounts in KRW billion. With 1–2 winners the median prize is 7.39 bn; with 26+ winners it is only 0.84 bn.

4. Have winner counts and prizes changed over time?

Question. From the first draws in 2002 to 2026, did the number of winners go up, and did the prize per winner stay the same?

What this calculation does. Spearman correlation measures whether a quantity rises or falls as the round number increases. A 52-round moving average smooths short-term noise. The data are also split into four eras so mean winners and median prizes can be compared side by side. A box plot shows the full spread of winners inside each era.

Analysis	Statistic	p-value
Spearman ρ (round number, winners)	+0.642	< 0.001
Spearman ρ (round number, prize per winner)	–0.071	0.013
Spearman ρ (round number, total pool)	+0.831	< 0.001

Winners and total pool rose strongly. Median prize per winner stayed almost flat near 2 bn KRW.

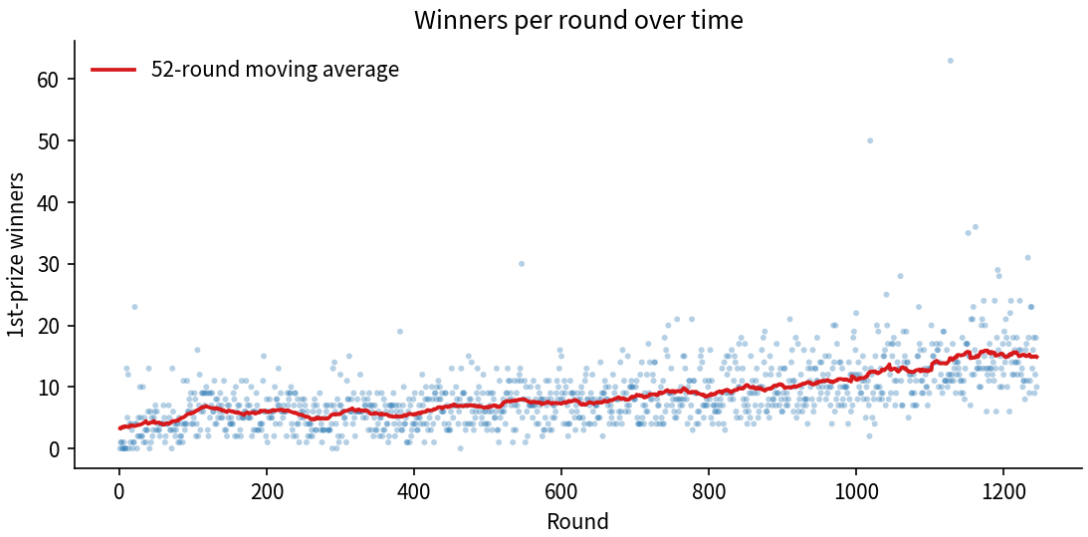


Figure 3. Winners per round and 52-round moving average. Mean winners: 5.2 (early) → 13.9 (recent).

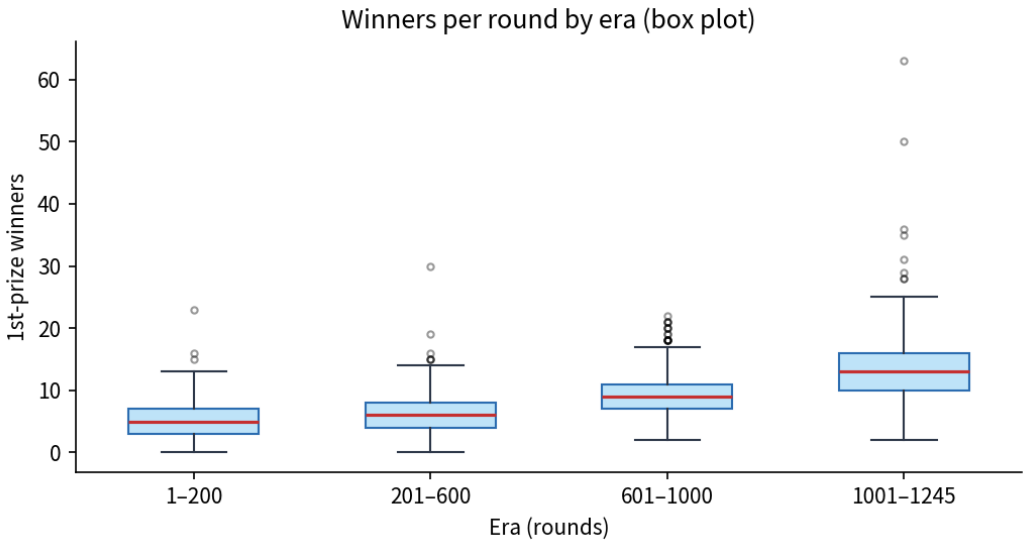


Figure 4. Box plot of winners by era. The box shows the middle 50 % of rounds; the line inside is the median; dots are extreme rounds.

Era	N	Mean winners	Median prize	Median total
1–200	200	5.17	2.71	13.36
201–600	400	6.36	1.89	11.26
601–1000	400	9.26	2.11	18.21
1001–1245	245	13.94	2.12	27.09

Prize and total in KRW billion. Median total pool roughly doubled from the first era to the last.

5. Do certain number patterns get shared by more winners?

Question. When the six winning numbers add up to a small total, or contain many consecutive pairs, do more people usually share the 1st prize?

What this calculation does. First we remove the long-term rise in winners by dividing each round's winner count by a 52-round moving average. The result is called "relative winners" (value 1 = typical for that period). Then we measure the rank correlation between relative winners and four simple features of the draw: the sum of the six numbers, the number of consecutive pairs (e.g. 12-13), the count of odd numbers, and the count of numbers that are 31 or lower (the classic birthday range). A positive correlation means that feature tends to bring more co-winners; a negative correlation means fewer co-winners.

Analysis	Statistic	p-value
Spearman ρ (sum of six numbers, relative winners)	–0.090	0.0015
Spearman ρ (consecutive pairs, relative winners)	–0.086	0.0024
Spearman ρ (odd-number count, relative winners)	–0.014	0.622
Spearman ρ (count of numbers ≤ 31, relative winners)	+0.025	0.386
OLS slope (relative winners ~ count ≤ 31)	+0.017 (SE 0.013)	0.207

Higher sum and more consecutive pairs are linked to slightly fewer co-winners. The birthday-range effect is weak and not statistically clear. After adjusting for four tests (Bonferroni threshold 0.0125), sum and consecutive pairs stay significant.

Relative winners by sum of the six numbers (five equal groups)

Sum group	Rounds	Mean relative winners
Lowest sum (Q1)	256	1.068
Q2	243	1.026
Q3	244	1.033
Q4	253	0.978
Highest sum (Q5)	235	0.936

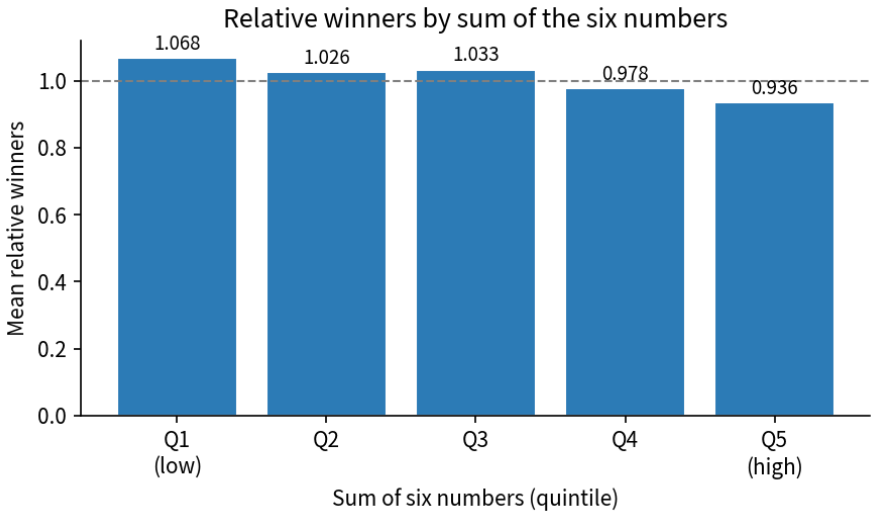


Figure 5. Draws with the highest number-sum have about 12–15 % fewer relative winners than draws with the lowest sum. A low sum usually means many small numbers — players seem to prefer small numbers overall.

Single numbers. No individual number shows a reliably higher or lower sharing rate after correction for 45 tests. The observed range of mean relative winners is only 0.93–1.13.

6. Extreme draws

Question. What were the biggest single prizes, the most shared jackpots, and the rounds with no winner at all?

What this section shows. Simple ranked lists of the largest prizes per winner and the rounds with the most winners. Zero-winner rounds are listed because the prize rolls over and the next draw often pays a higher amount.

Record	Value	Round
Largest prize for one winner	40.72 bn KRW	19
Most winners in one round	63 people	1128
Rounds with no 1st-prize winner	14 rounds	1,4,5,7,8,9,13,18,24,41,71,289,295,463

After a zero-winner round the median prize in the next draw was 6.43 bn KRW (11 cases), versus the usual median of 2.10 bn.

Five largest prizes per winner

Round	Winners	Prize (bn KRW)
19	1	40.72
25	2	24.23
20	1	19.35
43	1	17.75
15	1	17.01

Five rounds with the most winners

Round	Winners	Prize each (bn KRW)
1128	63	0.42
1019	50	0.44
1162	36	0.82
1152	35	0.87
1233	31	0.84

7. Why this analysis is useful

You cannot predict the next winning numbers. Every combination has the same chance. What you **can** know is how the money is shared and how the game has changed.

Sharing matters. If only one or two people win, each can receive around 7 billion KRW. If 25 or more people win, each may get less than 1 billion. The slope −0.68 puts a clear number on that drop.

More winners now. Twenty years ago a typical draw had about 5 winners. Recently it has about 14. The total prize money grew, but it is split among more people, so the amount per winner stayed near 2 billion KRW.

Number patterns have only a small effect. Draws with many small numbers or few consecutive pairs tend to be shared a little more. The difference is roughly 10–15 %. It does not change your chance of winning; it only slightly changes how many others you may share with.

Hot and cold numbers are normal. Some numbers have appeared a few more times than others, but the difference is within ordinary random variation. No number is “due” or “lucky” in a way that beats chance.

8. Main results in brief

• Numbers 34, 13 and 27 have appeared most often; 9, 22 and 32 least often. The differences are not statistically significant.
• When the number of 1st-prize winners doubles, the prize each person gets falls by about 38 % (log-log slope −0.68).
• Average winners per round rose from 5.2 to 13.9. Median prize per winner stayed near 2 bn KRW.
• Draws with a high sum of numbers or more consecutive pairs have slightly fewer co-winners (about 12–15 % less at the extremes).
• Biggest single prize: 40.72 bn KRW (round 19). Most shared jackpot: 63 winners (round 1128).

All numbers and figures come from the script lotto_analysis.py and the cleaned file lotto_cleaned.csv.